



DIRECTORATE OF PUBLIC INSTRUCTIONS (COLLEGES) PUNJAB
Intermediate classes December Test 2025

Class: 1st Year

Subject: Mathematics

Section: _____

Name: _____

Time: 3 Hours

Total Marks: 100

Roll No.: _____

Q#1: Fill the bubble by choosing the correct answer.

20

Sr#	Questions	Bubble
1	If $z = \cos x + i \sin x$ then $ z ^2 = ?$ a) 0 b) 1 c) 2 d) 3	☐ A ☐ B ☐ C ☐ D
2	If polar form of a complex number is $r \cos \theta + i r \sin \theta$ then $\theta = ?$ a) $\tan^{-1} \frac{y}{x}$ b) $\tan^{-1} \frac{x}{y}$ c) $\tan \frac{y}{x}$ d) $\tan \frac{x}{y}$	☐ A ☐ B ☐ C ☐ D
3	Argument of $i + \sqrt{3}$ is a) 60° b) 45° c) 30° d) 90°	☐ A ☐ B ☐ C ☐ D
4	Point of intersection of $f(x) = 2x + 5$ and $g(x) = -x + 5$ is a point which satisfies a) $f(x)$ b) both a and b c) $g(x)$ d) None of these	☐ A ☐ B ☐ C ☐ D
5	Graph of quadratic equation will be a) Line b) Hyperbola c) Parabola d) None of these	☐ A ☐ B ☐ C ☐ D
6	Maximum number of roots of quadratic equation are a) 2 b) 3 c) 1 d) None of these	☐ A ☐ B ☐ C ☐ D
7	The distinct roots of $x^2 + x^{-2} = 2$ are a) $\pm 1, \pm i$ b) ± 2 c) $\pm i$ d) ± 1	☐ A ☐ B ☐ C ☐ D
8	The cofactor of 0 in matrix $A = \begin{pmatrix} 1 & 1 & -1 \\ 1 & 0 & 1 \\ 2 & 1 & 1 \end{pmatrix}$ is a) 0 b) -1 c) -3 d) 3	☐ A ☐ B ☐ C ☐ D
9	If $o(A) = m \times n$, $o(B) = n \times p$, then $o(AB)^t$ is a) $m \times p$ b) $p \times m$ c) $m \times n$ d) $p \times n$	☐ A ☐ B ☐ C ☐ D
10	Transpose of a diagonal matrix is a) Scalar matrix b) Row matrix c) Diagonal matrix d) Null vector	☐ A ☐ B ☐ C ☐ D
11	The conditional equation $5x = 4$ is true if $x = ?$	☐ A ☐ B ☐ C ☐ D



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	a) 4 b) $5/4$	c) 5 d) $4/5$	
12	$\frac{P(x)}{x^2+1}$ will be proper fraction if degree of P(x) equals a) 1 b) 2 c) 3 d) 4		(A) (B) (C) (D)
13	The 9 th term of sequence 1, -3, 5, -7, ... is a) 15 b) 17 c) -15 d) -17		(A) (B) (C) (D)
14	No term of a harmonic sequence can be a) 2 b) 1 c) all given options are wrong d) 0		(A) (B) (C) (D)
15	Solution of $\sqrt{y+3} = \sqrt{3y-5}$ is a) 2 b) -4 c) 4 d) 1		(A) (B) (C) (D)
16	Distance between A(3,8) and B(5,6) is a) $\sqrt{2}$ b) 2 c) $2\sqrt{2}$ d) $1/2$		(A) (B) (C) (D)
17	$\cos 75^\circ =$ a) $\frac{\sqrt{3}-1}{2\sqrt{2}}$ b) $\frac{-\sqrt{3}+1}{2\sqrt{2}}$ c) $\frac{\sqrt{3}+1}{2\sqrt{2}}$ d) $\frac{-\sqrt{3}-1}{2\sqrt{2}}$		(A) (B) (C) (D)
18	$\sin 196^\circ = ?$ a) $\sin 16^\circ$ b) $\cos 16^\circ$ c) $-\sin 16^\circ$ d) $-\cos 16^\circ$		(A) (B) (C) (D)
19	Period of $2\cos 3x$ is a) π b) 2π c) $3\pi/2$ d) $2\pi/3$		(A) (B) (C) (D)
20	Trigonometric functions are a) Discontinuous b) Periodic c) Not Periodic d) None of these		(A) (B) (C) (D)



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Q# 2 Attempt any 8 short questions.

(8 × 2 = 16)

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| <ol style="list-style-type: none">1. Prove that $z = \bar{z}$ iff z is real.2. Find the square root of $z = -7 - 24i$.3. Factorize $2z^2 - 22z + 65$.4. Solve $(-1 + \sqrt{-3})^5 + (-1 - \sqrt{-3})^5$.5. Find point of intersection of coordinate axes and $y = 2x - 1$.6. Find the inverse of $f(x) = 2x^2 - 8x + 11$ for $x \geq 2$, and its domain and range. | <ol style="list-style-type: none">7. If $f(x) = \sin x$ then find $f(a+h) - f(a)$.8. Find domain and range of $g(x) = x-5$.9. Find maximum or minimum value of $f(x) = -x^2 + 8x + 13$.10. Solve $x^2 - 3x + 2 > 4$.11. Solve $\frac{1}{x+1} + \frac{2}{x+2} = \frac{7}{x+5}$, $x \neq -1, -2, -5$.12. Find 4th root of 625. |
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Q# 3 Attempt any 8 short questions.

(8 × 2 = 16)

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| <ol style="list-style-type: none">1. Define and give an example of matrix.2. Show that in general $(A+B)^2 \neq A^2 + B^2 + 2AB$.3. Without expansion show that $\begin{vmatrix} x & a+x & b+c \\ x & b+x & a+c \\ x & c+x & a+b \end{vmatrix} = 0$4. Define the echelon form of the matrix.5. Define improper rational fraction.6. Resolve $\frac{2}{x^2-1}$ into partial fractions. | <ol style="list-style-type: none">7. Without expansion show that $\begin{vmatrix} 1 & a & a^2-bc \\ 1 & b & b^2-ac \\ 1 & c & c^2-ab \end{vmatrix} = 0$8. Check if $\sin^4 x + \cos^4 x$ is even or odd function.9. Find the period of $19 \sin\left(\frac{\pi x}{20}\right)$.10. Determine whether $\sin^2 x$ is even or odd function.11. Find maximum and minimum values of $3 + \sin 2x$.12. What is meant by rank of a matrix. |
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Q# 4 Attempt any 9 short questions.

(9 × 2 = 18)

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| <ol style="list-style-type: none">1. Define sequence of real numbers.2. Write the n^{th} term of the sequence $\frac{1}{a_1}, \frac{1}{a_1+d}, \frac{1}{a_1+2d}, \dots$3. Find the 12th term of A.P. 8, 3, -2, ...4. Find 3 A.M.s. between $\sqrt{2}$ and $3\sqrt{2}$.5. Find 9th term of H.P. $\frac{1}{3}, \frac{1}{5}, \frac{1}{7}, \dots$6. Solve $\sin 810^\circ \sin 630^\circ + \cos 135^\circ \sin 225^\circ$.7. Prove that for a triangle ABC, $\sec\left(\frac{\alpha+\beta}{2}\right) = \operatorname{cosec} \frac{\gamma}{2}$ | <ol style="list-style-type: none">8. Find the value of $\tan 105^\circ$.9. Prove that $\frac{1 - \tan \theta \tan \phi}{1 + \tan \theta \tan \phi} = \frac{\cos(\theta + \phi)}{\cos(\theta - \phi)}$10. Prove that $1 + \tan x \tan 2x = \sec 2x$.11. Prove that $\frac{\sin 3x}{\sin x} - \frac{\cos 3x}{\cos x} = 2$.12. Express $\cos(x+y)\sin(x-y)$ as sum or difference.13. Solve $\frac{\sin 3x - \sin x}{\cos x - \cos 3x}$ |
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LONG QUESTIONS

Attempt any 3 long questions.

(6 × 5 = 30)

Q 5	(a) Find square root of $z = 119 + 120i$ and represent it on Argand Diagram. (b) Let $f : R \rightarrow R$ defined by $f(x) = \frac{2x-3}{x+1}$. Find domain and range of it and determine whether $f(x)$ is onto.
Q 6	(a) Solve $3x^2 + 15x - 2\sqrt{x^2 + 5x + 1} = 2$. (b) Find the inverse of the matrix $A = \begin{pmatrix} 1 & 1 & -1 \\ 1 & 0 & 1 \\ 2 & 1 & 1 \end{pmatrix}$.
Q 7	(a) Resolve into partial fractions $\frac{2x+32}{(x-2)(x^2+2)^2}$. (b) Find the sum of all integers in the first 1000 positive integers which are neither divisible by 2 nor by 5.
Q 8	(a) A man on the top of a 100m high light-house is in line with the two ships on the same side of it, whose angles of depression from the man are 17° and 19° respectively. Find the distance between ships. (b) Find the value of $\sin 10^\circ \sin 30^\circ \sin 50^\circ \sin 70^\circ$.
Q 9	(a) Solve $\cos^2\left(\frac{\pi}{4} + \frac{\alpha}{2}\right) - \cos^2\left(\frac{\pi}{4} - \frac{\alpha}{2}\right)$. (b) If $a^2, b^2, \&c^2$ are in A.P., show that $a+b, c+a, \&b+c$ are in H.P.